Bayesian Inference

Advanced Statistics Lab

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Theme

Bayes Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

In statistical modelling:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

Bayesian vs Frequentist

This is in contrast with a frequentist (maximum likelihood) approach, in which we aim to estimate non-random, fixed parameters using the likelihood of the data given those parameters only ($p(y|\theta)$).

The overall likelihood of the data ($p(y)$) only serves as a scaling constant:

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

Theme

Estimating A Normal Mean

In a Bayesian framework, we formulate our problem in terms of the assumed distribution of our data (likelihood) and the prior distributions of our parameters we wish to estimate.

$$\mathbf{y} \sim \mathbf{N}(\mu, \sigma_e^2)$$
$$\mu \sim N(\mu_{prior}, \sigma^2_{prior})$$

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brms Mean Example

```r
library(brms)
library(ggplot2)
library(knitr)
library(tidyverse)

data(iris)

# this gives us the default brms priors
get_prior(Sepal.Width ~ 1, data = iris)

# set a custom prior for the intercept
priors = set_prior("normal(0,0.2)", class = "Intercept")
```

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Fit The Model

```r
fit <- brm(Sepal.Width ~ 1,
      data = iris,
      family = gaussian(),
      prior = priors,
      sample_prior = TRUE,
      silent = 2,
      refresh = 0
)
```

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Prior And Posterior Samples

```
draws = as_draws_array(fit)[,1,, drop = TRUE]

samples = rbind(
    data.frame(type = "prior",
               samples = as.numeric(draws[, "prior_Intercept"])),
    data.frame(type = "posterior",
               samples = as.numeric(draws[, "b_Intercept"]))
)

kable(
  samples %>% group_by(type) %>% summarize(
    n = n(),
    mean = mean(samples),
    sd = sd(samples)
  ),
  digits = 3
)
```

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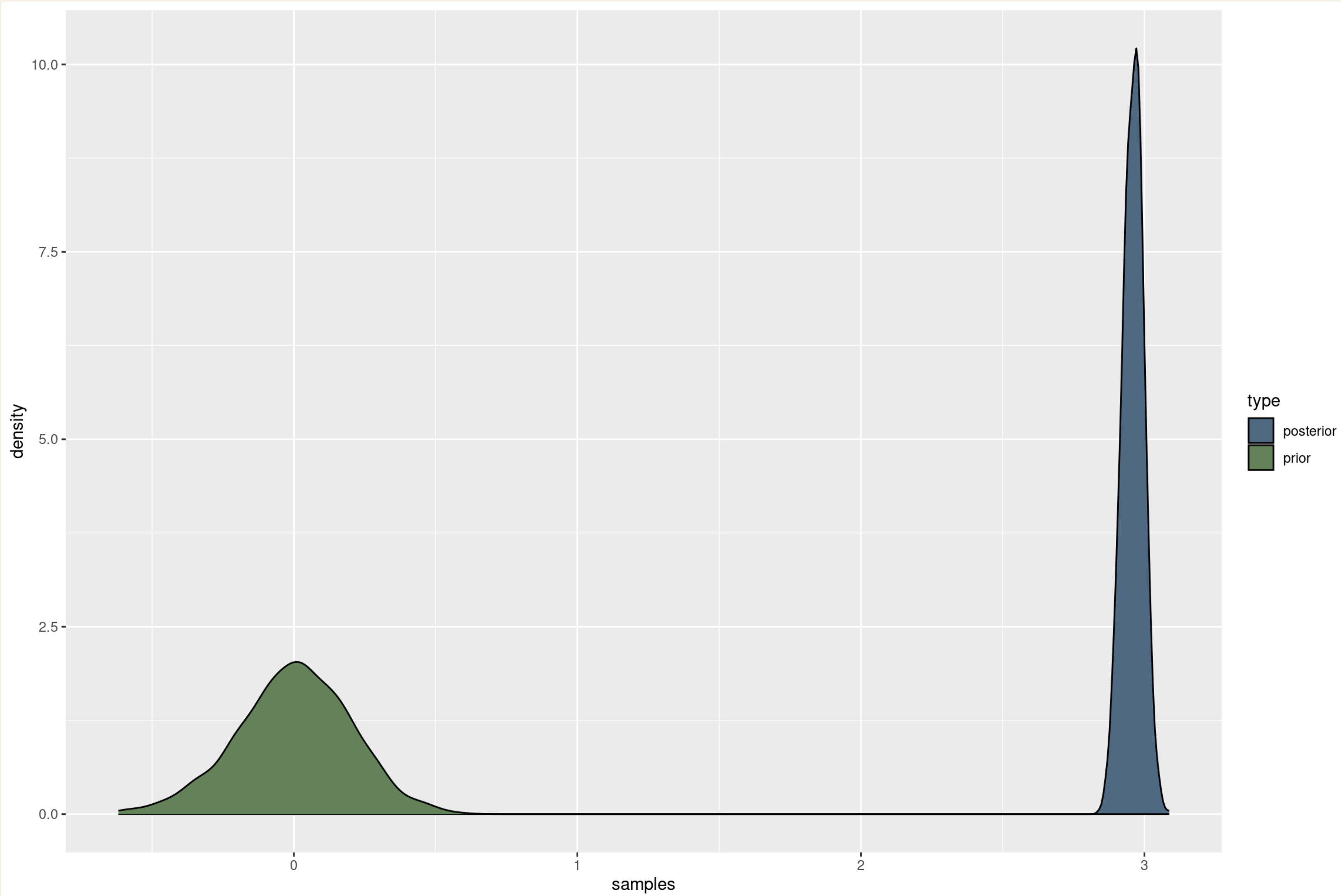
Prior And Posterior



Lab notes

Updating Priors From Existing Evidence

Bayesian Inference allows us to incorporate actual prior evidence in the estimation of parameters.

We might have conducted a previous experiment and arrived at an estimate of the mean of our observations.

Then we do another experiment of similar setup and want to update our mean estimate.

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Prior Updating

$$p(\theta|y)_{FirstExperiment} \propto p(y|\theta)p(\theta)$$

Plugging in the posterior from the first experiment as the prior for the second:

$$p(\theta|y)_{SecondExperiment} \propto p(y|\theta)p(\theta|y)_{FirstExperiment}$$

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Update Prior In R

```r
iris_first = iris[1:50,]

priors = set_prior("normal(0,4)", class = "Intercept")

fit_first <- brm(Sepal.Width ~ 1,
     data = iris_first,
     family = gaussian(),
     prior = priors,
     sample_prior = TRUE,
     silent = 2,
     refresh = 0
)

draws_first = as_draws_array(fit_first)[,1,, drop = TRUE]
```

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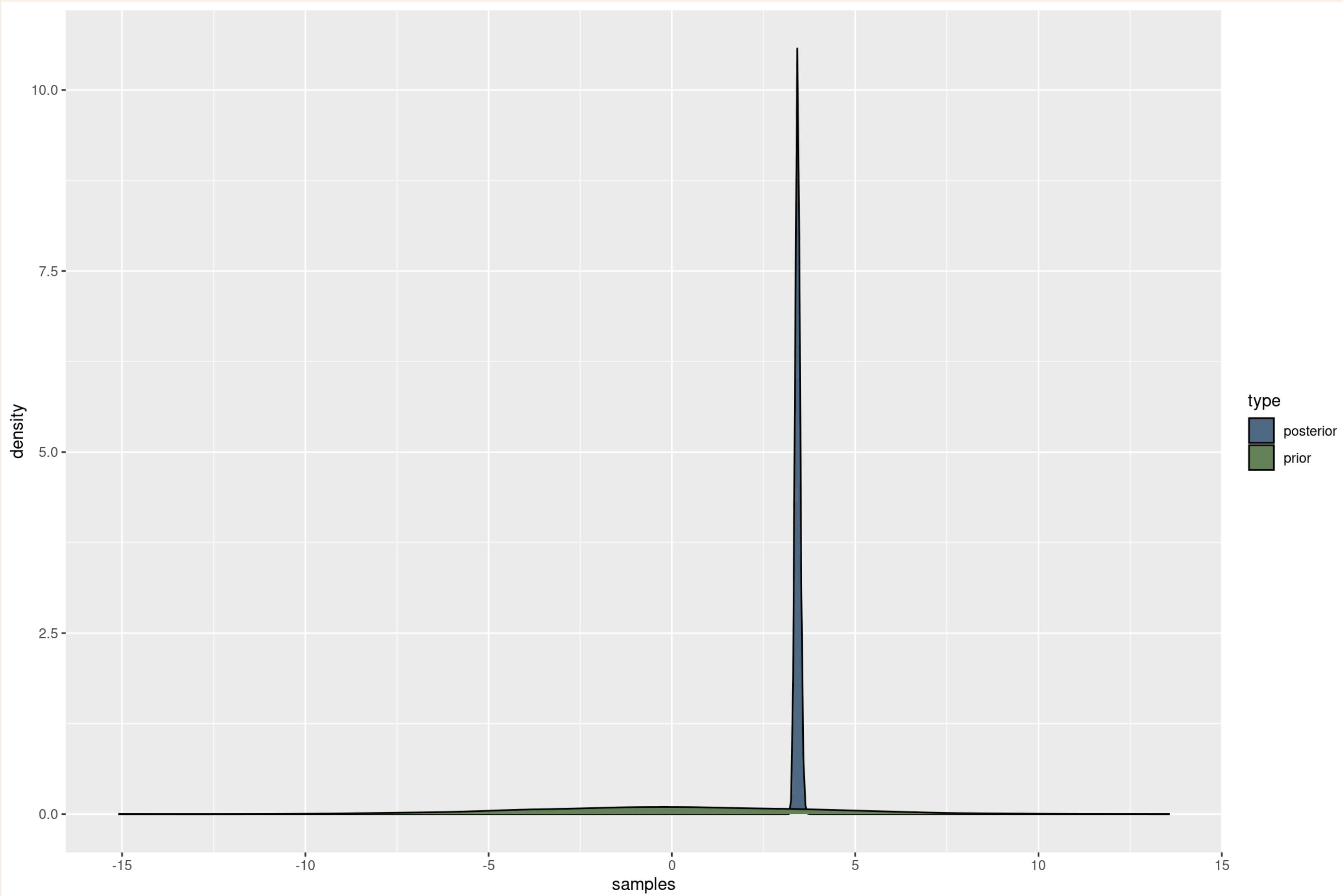
Use Posterior As Prior

```
samples_first = rbind(
  data.frame(type = "prior",
             samples = as.numeric(draws_first[, "prior_Intercept"])),
  data.frame(type = "posterior",
             samples = as.numeric(draws_first[, "b_Intercept"]))
)

mu_prior = mean(samples_first$samples[samples_first$type == "posterior"])
sd_prior = sd(samples_first$samples[samples_first$type == "posterior"])

priors = set_prior(
  paste("normal(", mu_prior, ",", sd_prior, ")", sep = ""),
  class = "Intercept"
)
```

First Experiment



Lab notes

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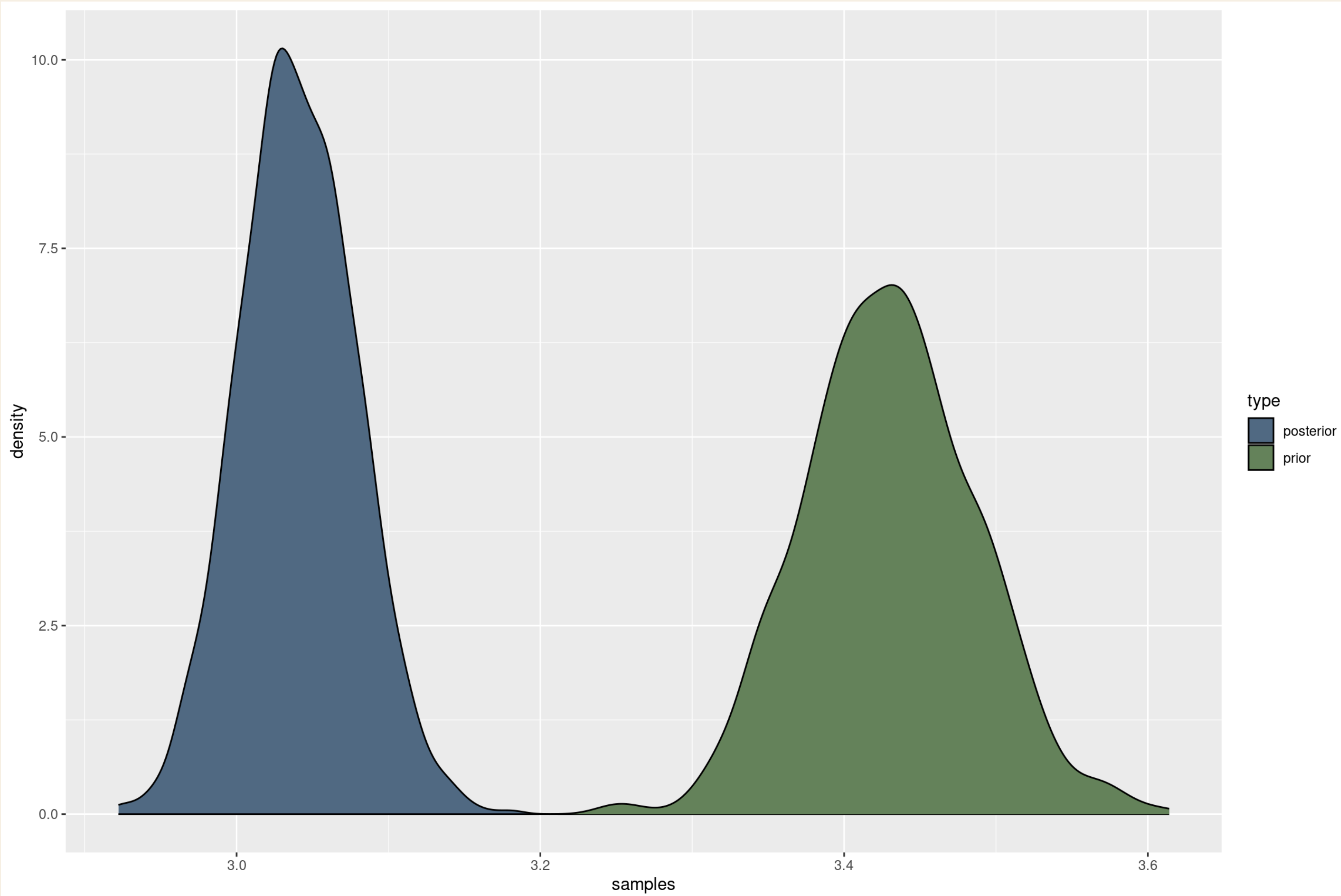
Second Experiment



Lab notes

Multivariate Animal Model In brms

$$\mathbf{Y} = \mathbf{XB} + \mathbf{ZU} + \mathbf{E}$$

Likelihood:

$$\mathbf{Y} \sim \mathbf{MVN}(\mathbf{XB} + \mathbf{ZU}, \mathbf{\Sigma}_E)$$

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Priors

$$
\begin{aligned}
\mathbf{B} &\sim \mathbf{U}(-Inf, Inf) \\
\mathbf{U} &\sim \mathbf{MVN}(\mathbf{0}, \mathbf{\Sigma}_U) \\
\mathbf{\Sigma}_U &= \mathbf{diag(S)}_U \mathbf{\Omega}_U \mathbf{diag(S)}_U \\
\mathbf{S}_U &\sim \mathbf{Cauchy}(0, 5) \\
\mathbf{\Omega}_U &\sim \mathbf{LKJ}(1)
\end{aligned}
$$

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brms Animal Model Formula

```
brmf <- brmsformula(
  mvbind(
    milk,
    fat
  ) ~ 1 +
  factor(lact) +
  herd +
  (1|g|gr(idA, cov = A)) + # additive genetic effect
  (1|p|gr(id)), # PE effect
  data = D,
  data2 = list(A = A)
)
```

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Check Stan Code

```
stanCode <- make_stancode(
  formula = brmf,
  data = D,
  data2 = list(A = A),
  family = gaussian()
)

stanData <- make_standata(
  formula = brmf,
  data = D,
  data2 = list(A = A),
  family = gaussian()
)

stanCode
```

Resources

Stan

brms

https://github.com/cheuerde/AnimalModels

https://github.com/cheuerde/BayesianAlphabet